

Introduction to Basic and Advanced Statistical Modelling

From LM to GLM

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Linear Models (LM)

LM — what is a statistical model?

Statistical model

an approximate description of the *data generating process*, i.e. an approximation of how the world generates data

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In particular, it helps

- correctly interpreting model parameters
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Interpreting coefficients — introduction

Interpreting the meaning of *regression coefficients* (= *parameter estimates*) correctly in statistical models is the most critical part given that:

- coefficients mediate the relationship between the statistics and the biology

Interpreting coefficients — data preparation: clean up

Let's create a single clean dataset with all the variables we will need for all models

```
penguins |>
  as_tibble() |>
  select(-starts_with("bill")) |>
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Generalised Linear Models (GLM)

GLM — why the need?

Classic linear models (LM) are useful to fit many phenomena, including many for which the relationship between a predictor and the response is not linear (e.g. polynomials), but they share the limitation that they are designed to approximate a data generating process that is intrinsically Gaussian (remember that the error is assumed normally distributed)

LM won't work well for:

- binary data (e.g. the presence/absence of a species)
- proportion (e.g. the hatching success in a nest), especially if values get close to 0 or 1
- count data (e.g. the number of species caught on camera), especially if counts are low
- time to event data (e.g. time to dispersion), especially if times are short

For these situations GLM are the first choice!

NB: logistic regression & poisson regressions are examples of GLM

GLM — notations of a GLM: simple notation

$$\eta_i = \beta_0 + \beta_1 \times x_{1,i} + \beta_2 \times x_{2,i} + \dots + \beta_p \times x_{p,i}$$

with

- $E(Y) = \mu = g^{-1}(\eta)$

- $\text{Var}(Y) = \phi V(\mu)$

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Tests

Tests — main testing strategies

Context	Frequentist approach	Bayesian approach
compare non-nested (not a <i>proper</i> test)	AIC / BIC	BF / LOO-CV or WAIC comparison
compare two nested models (general)	F-test (for LM) / LRT (for GLM*)	BF / LOO-CV or WAIC comparison / PIP
special case: model vs null model	F-test (for LM) / LRT (for GLM*) of full vs intercept-only	BF / LOO-CV or WAIC comparison / PIP
special case: overall effect of a variable	F-test (for LM) / LRT (for GLM*) of full vs variable dropped	BF / LOO-CV or WAIC comparison / PIP
coefficient vs a fixed value (usually 0)	Wald t/z-test** (for LM/GLM*) on the coefficient	BF (Savage-Dickey) / credible interval / PIP

* GLM with quasi family (i.e. $\phi \neq 1$), should be dealt with as an LM

** Wald tests perform poorly for small samples or parameters near a boundary, and alternatives exist (e.g. parametric bootstrap)

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* GLM with quasi family (i.e. $\phi \neq 1$), should be dealt with as an LM

** Wald tests perform poorly for small samples or parameters near a boundary, and alternatives exist (e.g. parametric bootstrap)

Technical vocabulary:

- AIC = Akaike information criterion
- BIC = Bayesian information criterion
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Tests

Tests — main testing strategies

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compare non-nested (not a <i>proper</i> test)	AIC / BIC	BF / LOO-CV or WAIC comparison
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NB:

- whatever you do, the response variable data must be strictly identical (beware of values dropped automatically due to NAs in predictors)
- to reduce false positives, perform tests in the following order: full vs null → full vs variable dropped → coefficient vs a fixed value

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Model construction

Model construction — TODO

- TODO

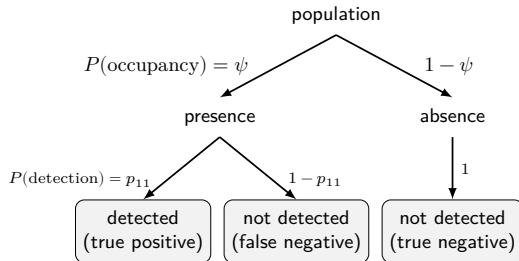
Assumptions

Assumptions — TODO

- TODO

Habitat occupancy model

Habitat occupancy model — recall the situation



We thus have a state space model (SSM):

- a non-observable layer with a state variable describing whether the habitat is occupied or not
- an observable outcome describing whether a particular species has been observed or not
- each of these levels can be modelled by a logistic GLM
- the 2 GLMs are not independent, it is a hierarchical model

Habitat occupancy model — fitting a no covariate model

We first simulate data and define the model

```
n_sites <- 100  
n_visits <- 50
```

Conclusions

Conclusions

- modelling can be tricky, especially when relying on MCMC
- simulating data is critical to make sure you do estimate what you think
- {nimble} is very flexible and can do much more than I showed you
- TODO
- TODO
- TODO

Questions?

Questions?

Developer corner

Ignore this slide — (technical info for Alex)

```
[1] "1.3 min of R running time"
```

```
R version 4.6.1 (2026-06-24)
```

```
Platform: x86_64-redhat-linux-gnu
```

```
Running under: Fedora Linux 44 (KDE Plasma Desktop Edition)
```

```
Matrix products: default
```

```
BLAS/LAPACK: FlexiBLAS OPENBLAS-OPENMP; LAPACK version 3.12.1
```

```
attached base packages:
```

```
[1] stats      graphics  grDevices datasets  utils      methods    base
```

```
other attached packages:
```

```
[1] modelbased_0.16.0 bayesplot_1.15.0 posterior_1.7.0 nimble_1.4.2 brms_2.23.0 Rcpp_1.1.2 lubridate_1.9.5  
[8] forcats_1.0.1 stringr_1.6.0 dplyr_1.2.1 purrr_1.2.2 readr_2.2.0 tidyr_1.3.2 tibble_3.3.1  
[15] ggplot2_4.0.3 tidyverse_2.0.0
```

```
loaded via a namespace (and not attached):
```

```
[1] Rdpack_2.6.6 gridExtra_2.3.1 inline_0.3.21 sandwich_3.1-3 rlang_1.3.0  
[6] magrittr_2.0.5 multcomp_1.4-31 otel_0.2.0 matrixStats_1.5.0 compiler_4.6.1  
[11] loo_2.10.1 vctr_0.7.3 reshape2_1.4.5 pkgconfig_2.0.3 crayon_1.5.3  
[16] fastmap_1.2.0 backports_1.5.1 labeling_0.4.3 utf8_1.2.6 CoprManager_0.5.8  
[21] rmarkdown_2.31 tzdb_0.5.0 pracma_2.4.6 nlptr_2.2.1 tinytex_0.60  
[26] bit_4.6.0 xfun_0.60 jsonlite_2.0.0 parallel_4.6.1 R6_2.6.1  
[31] stringi_1.8.9 RColorBrewer_1.1-3 StanHeaders_2.32.10 boot_1.3-32 numDeriv_2016.8-1.1  
[36] estimability_2.0.0 rstan_2.32.7 knitr_1.51 zoo_1.9-0 parameters_0.29.2  
[41] Matrix_1.7-6 splines_4.6.1 igraph_2.3.3 timechange_0.4.0 tidyselect_1.2.1  
[46] rstudioapi_0.19.0 abind_1.4-8 yaml_2.3.12 codetools_0.2-20 curl_7.1.0  
[51] pkgbuild_1.4.8 lattice_0.23-1 plyr_1.8.9 withr_3.0.3 bridgesampling_1.2-1  
[56] bayesplot_1.15.0 DT_2.8-0 codr_0.19-1 evaluate_1.0.5 marginaleffects_0.32.0  
[61] survival_3.8-9 RcppParallel_6.2.0 pillar_1.11.1 tensorA_0.36.2.1 checkmate_2.3.4
```